

HDOC - DAY 11

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SUBJECT: C programming

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Question 1

Find the sum of squares of the digits of a positive integer using a while loop. Example: Number = 123, Sum = 14.

1. Algorithm

1. Start the program.
2. Declare integer variables n, temp, digit and sum; initialize sum = 0.
3. Input a positive integer n.
4. Copy n into temp.
5. Repeat while temp > 0: find digit = temp % 10, add digit * digit to sum, and set temp = temp / 10.
6. Display the sum of squares of the digits.
7. Stop the program.

2. Test Case Table

Test Case	Input	Expected Output	Actual Output	Result
TC1	123	14	14	Pass
TC2	45	41	41	Pass
TC3	9	81	81	Pass
TC4	100	1	1	Pass
TC5	222	12	12	Pass

3. C Program

```
#include <stdio.h>
int main()
{
    int n, temp, digit, sum = 0;
    printf("Enter a positive integer: ");
    scanf("%d", &n);
    temp = n;
    while (temp > 0)
    {
        digit = temp % 10;
        sum = sum + digit * digit;
        temp = temp / 10;
    }
    printf("Sum of squares of digits = %d\n", sum);
    return 0;
}
```

4. Compilation, Execution and Testing

Write/Save: Save the above program as sumofsquares.

Compile: `$ gcc sumofsquares.c -lm`

Execute: `$ ./a.out`

```
(kali㉿kali)-[~/Desktop/cprog]
$ gcc sumofsquares.c -lm

(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
Enter a positive integer: 123
Sum of squares of digit= 14

(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
Enter a positive integer: 45
Sum of squares of digit= 41

(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
Enter a positive integer: 9
Sum of squares of digit= 81

(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
Enter a positive integer: 100
Sum of squares of digit= 1

(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
Enter a positive integer: 222
Sum of squares of digit= 12
```

Question 2

Find the sum of cubes of the digits of a positive integer using a for loop. Example: Number = 123, Sum = 36.

1. Algorithm

1. Start the program.
2. Declare integer variables n, temp, digit and sum; initialize sum = 0.
3. Input a positive integer n.
4. Copy n into temp.
5. Use a for loop while temp > 0. In each iteration, find digit = temp % 10, add digit * digit * digit to sum, and set temp = temp / 10.
6. Display the sum of cubes of the digits.
7. Stop the program.

2. Test Case Table

Test Case	Input	Expected Output	Actual Output	Result
TC1	123	36	36	Pass
TC2	45	189	189	Pass
TC3	9	729	729	Pass
TC4	100	1	1	Pass
TC5	222	24	24	Pass

3. C Program

```
#include <stdio.h>
int main()
{
    int n, temp, digit, sum = 0;
    printf("Enter a positive integer: ");
    scanf("%d", &n);
    temp = n;
    for (; temp > 0; temp = temp / 10)
    {
        digit = temp % 10;
        sum = sum + digit * digit * digit;
    }
    printf("Sum of cubes of digits = %d\n", sum);
    return 0;
}
```

4. Compilation, Execution and Testing

Write/Save: Save the above program as cube.

Compile: \$ gcc cube.c -lm

Execute: \$./a.out

```
(kali㉿kali)-[~/Desktop/cprog]
$ gcc cube.c -lm
```

```
(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
```

Enter a positive integer: 123

Sum of cubes of digits= 36

```
(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
```

Enter a positive integer: 45

Sum of cubes of digits= 189

```
(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
```

Enter a positive integer: 9

Sum of cubes of digits= 729

```
(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
```

Enter a positive integer: 100

Sum of cubes of digits= 1

```
(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
```

Enter a positive integer: 222

Sum of cubes of digits= 24

Question 3

Find the difference between the sum of cubes and the sum of squares of the digits of a positive integer using a do-while loop. Example: Number = 123, Difference = 22.

1. Algorithm

1. Start the program.
2. Declare integer variables n, temp, digit, squareSum and cubeSum; initialize both sums to 0.
3. Input a positive integer n and copy it into temp.
4. Using a do-while loop, extract each digit with temp % 10.
5. Add digit * digit to squareSum and digit * digit * digit to cubeSum.
6. Update temp = temp / 10 and repeat while temp > 0.
7. Calculate difference = cubeSum - squareSum.
8. Display the difference and stop.

2. Test Case Table

Test Case	Input	Expected Output	Actual Output	Result
TC1	123	22	22	Pass
TC2	45	148	148	Pass
TC3	9	648	648	Pass
TC4	100	0	0	Pass
TC5	222	12	12	Pass

3. C Program

```
#include <stdio.h>
int main()
{
    int n, temp, digit, squareSum = 0, cubeSum = 0, difference;
    printf("Enter a positive integer: ");
    scanf("%d", &n);
    temp = n;
    do
    {
        digit = temp % 10;
        squareSum = squareSum + digit * digit;
        cubeSum = cubeSum + digit * digit * digit;
        temp = temp / 10;
    } while (temp > 0);
    difference = cubeSum - squareSum;
    printf("Difference = %d\n", difference);
    return 0;
}
```

4. Compilation, Execution and Testing

Write/Save: Save the above program as difference.c.

Compile: \$ gcc difference.c -lm

Execute: \$./a.out

```
(kali㉿kali)-[~/Desktop/cprog]
$ gcc difference.c -lm
(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
Enter a positive integer: 123
Difference= 22
(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
Enter a positive integer: 45
Difference= 148
```

```
(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
Enter a positive integer: 9
Difference= 648
(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
Enter a positive integer: 100
Difference= 0
(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
Enter a positive integer: 222
Difference= 12
```